

Quiz 2 - Instructor key (revision 1)

Five one-page forms; 10 minutes; 3 points. Question 1 is easy, Question 2 is medium-hard, and Question 3 is hard on every form. Every item has different numerical inputs and a different correct value on all five forms.

Form	Q1	Q2	Q3	Item order
TW3R	E	C	A	Easy / Medium-hard / Hard
J2WJ	C	B	F	Easy / Medium-hard / Hard
XA44	F	G	E	Easy / Medium-hard / Hard
8NDK	G	E	D	Easy / Medium-hard / Hard
EA3R	D	A	B	Easy / Medium-hard / Hard

Form TW3R - worked solutions

Question 1 (easy) - E: 2

An i.i.d. sample of 100 workers has a sample standard deviation of hourly earnings of 20 euros per hour. What is the estimated standard error of the sample mean, in euros per hour?

The standard error is $s / \sqrt{n} = 20 / 10 = 2$ euros per hour. It estimates the sampling SD of the mean, not the dispersion of individual earnings.

Question 2 (medium-hard) - C: (-2; 5% only)

An i.i.d. sample has $n = 100$, sample mean $= 30$, and sample SD $= 50$. Test the population mean: $H_0: \mu = 40$ against $H_1: \mu \text{ is not } 40$. Use the large-sample normal approximation with two-sided critical values 1.96 (5%) and 2.576 (1%). Which pair gives (signed t-statistic; levels at which H_0 is rejected)?

$SE = 50 / \sqrt{100} = 5$. $t = (30 - 40) / 5 = -2$. Compare $|t| = 2$ with 1.96 and 2.576: reject at 5% only. The statistic remains signed, but both tails count for the decision. The normal reference is a large-sample approximation justified by the CLT and consistent estimation of the variance; the label t-statistic alone does not supply an exact Student t distribution.

Question 3 (hard) - A: (4; yes; no)

For $n > 1$, $Y_1, \dots, Y_n$ are i.i.d. with mean μ and variance 16. Let R be the mean of $Y_2, \dots, Y_n$ and define $T = (Y_1 + R)/2$. As n grows, which triple gives (limit of $\text{Var}(T)$; is T unbiased for μ ?; is T consistent for μ)?

The weights sum to one: $E(T) = (\mu + \mu)/2 = \mu$, so T is unbiased. Independence gives $\text{Var}(R) = 16/(n - 1)$ and $\text{Var}(T) = 16/4 + 16/[4(n - 1)]$, whose limit is 4. The LLN gives $R \rightarrow \mu$ in probability. Consequently, T approaches $(Y_1 + \mu)/2$ in probability, a nonconstant random variable because $\text{Var}(Y_1)$ is positive. It does not converge in probability to μ , so it is inconsistent. This argument uses the LLN and the nonconstant limit; a positive variance limit alone does not prove inconsistency for an arbitrary estimator sequence.

Form J2WJ - worked solutions

Question 1 (easy) - C: 3

An i.i.d. sample of 100 workers has a sample standard deviation of hourly earnings of 30 euros per hour. What is the estimated standard error of the sample mean, in euros per hour?

The standard error is $s / \sqrt{n} = 30 / 10 = 3$ euros per hour. It estimates the sampling SD of the mean, not the dispersion of individual earnings.

Question 2 (medium-hard) - B: (-2.2; 5% only)

An i.i.d. sample has $n = 144$, sample mean $= 34$, and sample SD $= 60$. Test the population mean: $H_0: \mu = 45$ against $H_1: \mu$ is not 45. Use the large-sample normal approximation with two-sided critical values 1.96 (5%) and 2.576 (1%). Which pair gives (signed t-statistic; levels at which H_0 is rejected)?

$SE = 60 / \sqrt{144} = 5$. $t = (34 - 45) / 5 = -2.2$. Compare $|t| = 2.2$ with 1.96 and 2.576: reject at 5% only. The statistic remains signed, but both tails count for the decision. The normal reference is a large-sample approximation justified by the CLT and consistent estimation of the variance; the label t-statistic alone does not supply an exact Student t distribution.

Question 3 (hard) - F: (5; yes; no)

For $n > 1$, $Y_1, \dots, Y_n$ are i.i.d. with mean μ and variance 20. Let R be the mean of $Y_2, \dots, Y_n$ and define $T = (Y_1 + R)/2$. As n grows, which triple gives (limit of $\text{Var}(T)$; is T unbiased for μ ?; is T consistent for μ)?

The weights sum to one: $E(T) = (\mu + \mu)/2 = \mu$, so T is unbiased. Independence gives $\text{Var}(R) = 20/(n - 1)$ and $\text{Var}(T) = 20/4 + 20/[4(n - 1)]$, whose limit is 5. The LLN gives $R \rightarrow \mu$ in probability. Consequently, T approaches $(Y_1 + \mu)/2$ in probability, a nonconstant random variable because $\text{Var}(Y_1)$ is positive. It does not converge in probability to μ , so it is inconsistent. This argument uses the LLN and the nonconstant limit; a positive variance limit alone does not prove inconsistency for an arbitrary estimator sequence.

Form XA44 - worked solutions

Question 1 (easy) - F: 4

An i.i.d. sample of 100 workers has a sample standard deviation of hourly earnings of 40 euros per hour. What is the estimated standard error of the sample mean, in euros per hour?

The standard error is $s / \sqrt{n} = 40 / 10 = 4$ euros per hour. It estimates the sampling SD of the mean, not the dispersion of individual earnings.

Question 2 (medium-hard) - G: (-2.4; 5% only)

An i.i.d. sample has $n = 196$, sample mean $= 38$, and sample SD $= 70$. Test the population mean: $H_0: \mu = 50$ against $H_1: \mu \text{ is not } 50$. Use the large-sample normal approximation with two-sided critical values 1.96 (5%) and 2.576 (1%). Which pair gives (signed t-statistic; levels at which H_0 is rejected)?

$SE = 70 / \sqrt{196} = 5$. $t = (38 - 50) / 5 = -2.4$. Compare $|t| = 2.4$ with 1.96 and 2.576: reject at 5% only. The statistic remains signed, but both tails count for the decision. The normal reference is a large-sample approximation justified by the CLT and consistent estimation of the variance; the label t-statistic alone does not supply an exact Student t distribution.

Question 3 (hard) - E: (6; yes; no)

For $n > 1$, $Y_1, \dots, Y_n$ are i.i.d. with mean μ and variance 24. Let R be the mean of $Y_2, \dots, Y_n$ and define $T = (Y_1 + R)/2$. As n grows, which triple gives (limit of $\text{Var}(T)$; is T unbiased for μ ?; is T consistent for μ)?

The weights sum to one: $E(T) = (\mu + \mu)/2 = \mu$, so T is unbiased. Independence gives $\text{Var}(R) = 24/(n - 1)$ and $\text{Var}(T) = 24/4 + 24/[4(n - 1)]$, whose limit is 6. The LLN gives $R \rightarrow \mu$ in probability. Consequently, T approaches $(Y_1 + \mu)/2$ in probability, a nonconstant random variable because $\text{Var}(Y_1)$ is positive. It does not converge in probability to μ , so it is inconsistent. This argument uses the LLN and the nonconstant limit; a positive variance limit alone does not prove inconsistency for an arbitrary estimator sequence.

Form 8NDK - worked solutions

Question 1 (easy) - G: 5

An i.i.d. sample of 100 workers has a sample standard deviation of hourly earnings of 50 euros per hour. What is the estimated standard error of the sample mean, in euros per hour?

The standard error is $s / \sqrt{n} = 50 / 10 = 5$ euros per hour. It estimates the sampling SD of the mean, not the dispersion of individual earnings.

Question 2 (medium-hard) - E: (-2.6; both 5% and 1%)

An i.i.d. sample has $n = 256$, sample mean $= 42$, and sample SD $= 80$. Test the population mean: $H_0: \mu = 55$ against $H_1: \mu$ is not 55. Use the large-sample normal approximation with two-sided critical values 1.96 (5%) and 2.576 (1%). Which pair gives (signed t-statistic; levels at which H_0 is rejected)?

$SE = 80 / \sqrt{256} = 5$. $t = (42 - 55) / 5 = -2.6$. Compare $|t| = 2.6$ with 1.96 and 2.576: reject at both 5% and 1%. The statistic remains signed, but both tails count for the decision. The normal reference is a large-sample approximation justified by the CLT and consistent estimation of the variance; the label t-statistic alone does not supply an exact Student t distribution.

Question 3 (hard) - D: (7; yes; no)

For $n > 1$, $Y_1, \dots, Y_n$ are i.i.d. with mean μ and variance 28. Let R be the mean of $Y_2, \dots, Y_n$ and define $T = (Y_1 + R)/2$. As n grows, which triple gives (limit of $\text{Var}(T)$; is T unbiased for μ ?; is T consistent for μ)?

The weights sum to one: $E(T) = (\mu + \mu)/2 = \mu$, so T is unbiased. Independence gives $\text{Var}(R) = 28/(n - 1)$ and $\text{Var}(T) = 28/4 + 28/[4(n - 1)]$, whose limit is 7. The LLN gives $R \rightarrow \mu$ in probability. Consequently, T approaches $(Y_1 + \mu)/2$ in probability, a nonconstant random variable because $\text{Var}(Y_1)$ is positive. It does not converge in probability to μ , so it is inconsistent. This argument uses the LLN and the nonconstant limit; a positive variance limit alone does not prove inconsistency for an arbitrary estimator sequence.

Form EA3R - worked solutions

Question 1 (easy) - D: 6

An i.i.d. sample of 100 workers has a sample standard deviation of hourly earnings of 60 euros per hour. What is the estimated standard error of the sample mean, in euros per hour?

The standard error is $s / \sqrt{n} = 60 / 10 = 6$ euros per hour. It estimates the sampling SD of the mean, not the dispersion of individual earnings.

Question 2 (medium-hard) - A: (-2.8; both 5% and 1%)

An i.i.d. sample has $n = 324$, sample mean = 46, and sample SD = 90. Test the population mean: $H_0: \mu = 60$ against $H_1: \mu \text{ is not } 60$. Use the large-sample normal approximation with two-sided critical values 1.96 (5%) and 2.576 (1%). Which pair gives (signed t-statistic; levels at which H_0 is rejected)?

$SE = 90 / \sqrt{324} = 5$. $t = (46 - 60) / 5 = -2.8$. Compare $|t| = 2.8$ with 1.96 and 2.576: reject at both 5% and 1%. The statistic remains signed, but both tails count for the decision. The normal reference is a large-sample approximation justified by the CLT and consistent estimation of the variance; the label t-statistic alone does not supply an exact Student t distribution.

Question 3 (hard) - B: (8; yes; no)

For $n > 1$, $Y_1, \dots, Y_n$ are i.i.d. with mean μ and variance 32. Let R be the mean of $Y_2, \dots, Y_n$ and define $T = (Y_1 + R)/2$. As n grows, which triple gives (limit of $\text{Var}(T)$; is T unbiased for μ ?; is T consistent for μ)?

The weights sum to one: $E(T) = (\mu + \mu)/2 = \mu$, so T is unbiased. Independence gives $\text{Var}(R) = 32/(n - 1)$ and $\text{Var}(T) = 32/4 + 32/[4(n - 1)]$, whose limit is 8. The LLN gives $R \rightarrow \mu$ in probability. Consequently, T approaches $(Y_1 + \mu)/2$ in probability, a nonconstant random variable because $\text{Var}(Y_1)$ is positive. It does not converge in probability to μ , so it is inconsistent. This argument uses the LLN and the nonconstant limit; a positive variance limit alone does not prove inconsistency for an arbitrary estimator sequence.